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Bioinspired bicipital muscle with fiber-constrained dielectric elastomer actuator

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ABSTRACT

Dielectric elastomers have become promising candidates for applications of artificial muscle, due to their outstanding properties on large deformation, fast response and high energy density. Diverse functional devices based on dielectric elastomer actuators have been developed. To mimic a slender shape muscle-like actuator with the capability of large actuation, however still remains a challenge. The fiber constrained dielectric elastomer actuator (FCDEA) was previously presented but only 30% unidirectional actuation strain was obtained. In this work, By using suitable stiff fibers to withstand large horizontal pre-stretch, up to 142% linear actuation strain is achieved. We demonstrate that the actuator is independent of the length ratio, making it possible to mimic natural muscle of slender shape. Theoretical predictions agree remarkably well with the experimental results. As a demonstration, we incorporate the FCDEA into a bioinspired artificial arm. After applying voltage, 70 degree rotation is achieved for the forearm relative to the fixed upper arm. The bioinspired design with the unique properties of dielectric elastomers shows the potential to use FCDEA to function as artificial muscle.

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1. Introduction

Technologies of artificial muscle have been rapidly developed in recent years. Diverse actuation mechanisms of functional materials include pneumatic [1,2], electrical [3,4], chemical [5,6], or thermal types [7,8]. Dielectric elastomer actuators (DEA), as one kind of electroactive polymer actuators, are particularly attractive because they produce fast and large deformation in response to an applied voltage, with comparable energy density of natural muscle [9,10]. Representative demonstrations of utilizing DEA to function as muscle include a bioinspired tunable lens [11], a walking robot [12], a worm-like robot [13], and an arm-wrestling robot [14].

A membrane of a dielectric elastomer expands in area and squeezes in thickness subject to a voltage through the thickness. The record of voltage-induced large deformation of dielectric elastomers keeps being updated from the 150% areal strain in the pioneering work to the recently achieved 2200% areal strain [15–17]. The voltage-induced deformation highly depends on how the mechanical loads and boundary conditions are applied [18]. In particular, achieving large unidirectional actuation strain to function as muscle remains a challenge. Up to 380% linear actuation strain is achieved on the DEA with the pure-shear configuration [19]. A recent improved design replaced the dead force with a passive membrane, making the design more compact [20]. However this pure shear actuator fails in the analogy of slender shape of natural muscle and the considerable mass of the clamped splint limits its application. To solve the above issues, fiber constrained dielectric elastomer actuator (FCDEA) was developed but only 30% actuation strain was obtained [18,21]. In this work, significant improvement in linear actuation is achieved by applying suitable stiff fibers to withstand large

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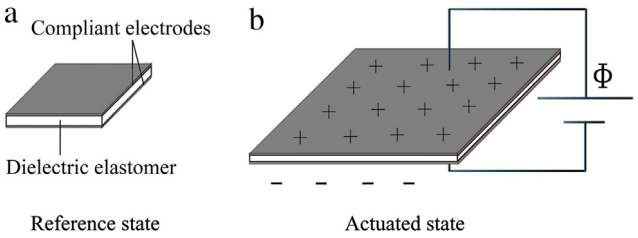


Fig. 1. A membrane of a dielectric elastomer deforms subject to a voltage through the thickness. (a) Reference state. (b) Actuated state.

pre-stretch. 140% linear unidirectional actuation strain is obtained and the actuation is independent of the shape, making the presented FCDEA promising for applications of artificial muscle. As a demonstration, we design a structure actuated by the FCDEA to mimic the function of bicipital muscle. By applying a voltage, the forearm can rotate 70° relative to the fixed upper arm. The bioinspired design with the unique properties of dielectric elastomers shows the potential to use FCDEA to function as artificial muscle.

2. Fiber-constrained dielectric elastomer actuator (FCDEA)

2.1. Design of FCDEA

A free standing rectangular membrane of a dielectric elastomer deforms biaxially subject to a voltage, as shown in Fig. 1. If one direction of the membrane is constrained, for example by a splint, the voltage induced deformation becomes unidirectional. Moreover, if one direction of the membrane is pre-stretched the membrane becomes anisotropic such that the other direction (normal to the pre-stretch) is much easier to be actuated. In principle, if the stretch state is far from the limiting stretch the larger of the horizontal pre-stretch, the larger vertical actuation strain can be achieved. Therefore, when fibers are horizontally embedded in a horizontally pre-stretched membrane, the large vertical actuation strain is expected to be achievable, as shown in Fig. 2. The membrane is subject to a vertical dead force as well. The local amplification of a segment of FCDEA is like the pure-shear actuator where the horizontal deformation is constrained by a splint, as shown in Fig. 2. The FCDEA can be viewed as a series of pure-shear actuators.

The key challenge of the present design is whether the fibers can firmly constrain the membrane or not. In order to sustain large horizontal pre-stretch, fibers should withstand large compressive stress. We can estimate the material and geometric parameters to prevent buckling of fibers. The critical buckling stress is $\sigma = \pi^2 Y d^2/\left(16 l^2\right)$, where d is the section diameter, l is the length, Y is the Young's modulus of each fiber. The true stress σ balances with the elasticity of the pre-stretched elastomer. The adhesion between fibers and the elastomer also contributes to prevent buckling, which is however difficult to control. The fibers we use in the following experiment are of diameter 0.3 mm and length 50 mm. The critical buckling stress is approximately $\sigma = Y \times 10^{-5}$. The Young's modulus of Nylon fibers is in the order of 10^9 Pa, and thus the maximum stress that can be sustained without buckling is about 10^4 Pa. The shear modulus of dielectric elastomer we use in the experiment is in the order of 10^4 Pa. Therefore, A large pre-stretch will induce dramatic buckling and thus lose functionality [20] (see Section 2.3 for the equations of force balance). If we use metallic fibers with Young's modulus about 10^{11} Pa, the buckling of fiber can be avoided but the actuator becomes too massive and the overall energy density becomes much lower. In the present experiment we use glass fibers with Young's modulus about 10^{10} Pa, making the actuator exhibit an excellent performance.

2.2. Experiment

The fabrication procedure of FCDEA is similar to that described in Ref. [18]. A membrane of a dielectric elastomer (3M VHB 4905) with initial thickness 0.5 mm is first pulled horizontally by 5 times and then the left and right sides are clamped to a rigid frame. The total length of the pre-stretched membrane is about 150 mm. Glass fibers with diameter 0.3 mm and length 50 mm are placed parallel, with a spacing of 1 mm, on the middle part of the membrane. Another layer of VHB 4905 membrane is pre-stretched in the same way and pushed down to stick to the previous one. Glass fibers are sandwiched between the two membranes. Carbon grease (ET-120C. Nascent) is brushed on the surfaces of the fiber-reinforced membrane with a border of 2 mm in the left and right side. Plastic plates are glued at the top and bottom of the membrane. The effective part of the composite membrane (with fibers embedded) is cut from the frame. Due to the horizontal constrains, the length of the membrane stays equal to the fiber length 50 mm without buckling. Then the membrane is stretched along the vertical direction by a constant force $P = 1.2$ N. The composite membrane is relaxed for 30 min until no obvious deformation can be observed. The vertical length at this rest state is 62 mm, and the vertical pre-stretch is about 1.5. A voltage with a ramping rate of 10 V/s generated by a waveform generator (Agilent 33220A) and a high voltage amplifier (Trek 610E) is applied to the membrane through the thickness. The deformation of the actuator is recorded by a digital camera until electrical breakdown occurs. The experiment is done at temperature 14 °C and humidity 38%.

Under the voltage of 6.3 kV the actuator suffers electrical breakdown and a hole forms in the middle of the membrane. Wrinkles are observed very near electrical breakdown and at that time the actuator attains its maximum length 150 mm. The voltage-induced linear unidirectional actuation strain is 150 mm/62 mm − 1 = 142%, as shown in Fig. 3. Significant improvement

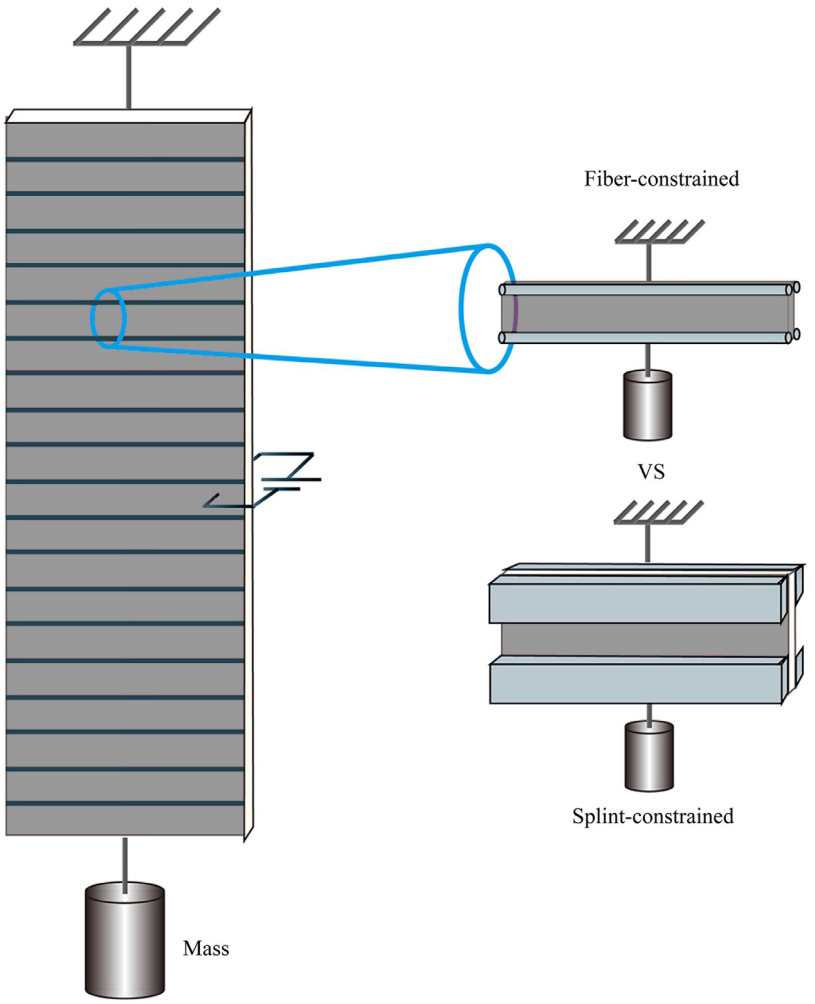


Fig. 2. Schematic of a fiber constrained dielectric elastomer actuator in which fibers constrain the horizontal deformation. The amplification of the local segment shows the similarity to the pure shear configuration with clamped splints.

is achieved compared to that in Ref. [18] where Nylon fibers were susceptible to buckling. Table 1 shows the collected data of unidirectional actuation strain of dielectric elastomer actuators using fiber constraints from previous works [18,21–23]. The presented FCDEA exhibits an obvious advantage in large actuation. Optimization of fiber material, fiber spacing and fiber orientation may further improve the actuation strain. Fig. 4 demonstrates the independence of the actuation strain on the length of actuator. For different lengths of actuators 15 mm, 28 mm, 58 mm at rest state, the actuated length becomes 34 mm, 65 mm, 134 mm respectively, with the average actuation strain 130%. This enables FCDEA to be capable of producing both large actuation strain and large actuation displacement. As we expected, larger horizontal pre-stretch results in larger actuation strain along vertical direction, as shown in Fig. 5.

2.3. Theoretical modeling

For a rectangular membrane, define stretches as the current length at actuated state divided by the reference length $\lambda_1 = l_1/L_1$, $\lambda_2 = l_2/L_2$, where 1 denotes the horizontal direction and 2 denotes the vertical direction. The true stresses σ_1, σ_2 are defined as the applied mechanical forces divided by the current section areas $\sigma_1 = P_1/(l_1 h)$ and $\sigma_2 = P_2/(l_2 h)$. The elastomer is taken to be incompressible such that the current thickness h relates the reference thickness H by $h = H/(\lambda_1\lambda_2)$. When a voltage Φ is applied to the membrane, the electric field is $E = \Phi/h$. The equilibrium equations for an ideal dielectric elastomer subject to both mechanical force and voltage are [24]

$$\sigma_1 + \varepsilon E^2 = \lambda_1 \frac{\partial W_{\text{stretch}}(\lambda_1, \lambda_2)}{\partial \lambda_1}, \tag{1}$$

$$\sigma_2 + \varepsilon E^2 = \lambda_2 \frac{\partial W_{\text{stretch}}(\lambda_1, \lambda_2)}{\partial \lambda_2}, \tag{2}$$

where $W_{\text{stretch}}(\lambda_1, \lambda_2)$ is the Helmholtz free energy associated with stretching the elastomer. In the model of ideal dielectric elastomer, the permittivity ε is taken as a constant independent of deformation. Eqs. (1), (2) show

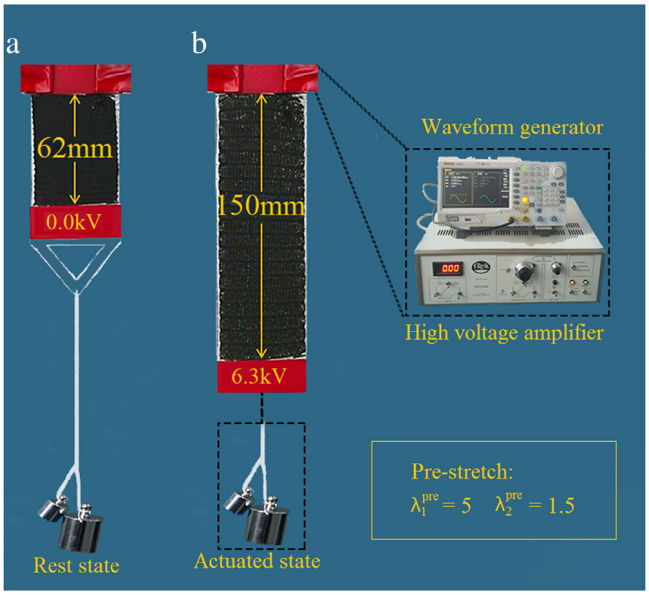


Fig. 3. Experimental images of FCDEA. (a) Rest state. (b) Actuated state at 6.3 kV, 142% unidirectional actuation strain is obtained.

the elasticity of the elastomer is balanced by the applied stresses along with the electrostatic interaction εE^2.

During deformation of the FCDEA, the horizontal stretch is fixed by the fiber $\lambda_1 = \lambda_1^{\text{pre}}$. In the vertical direction, the applied force P is a constant. Therefore, Eqs. (1), (2) become

$$\lambda_2 \frac{P}{L_1 H} + \varepsilon \left(\lambda_1^{\text{pre}} \lambda_2 \frac{\Phi}{H}\right)^2 = \frac{\mu \left(\lambda_2^2 - \left(\lambda_1^{\text{pre}} \lambda_2\right)^{-2}\right)}{1 - \left(\left(\lambda_1^{\text{pre}}\right)^2 + \lambda_2^2 + \left(\lambda_1^{\text{pre}} \lambda_2\right)^{-2} - 3\right)/J_{\text{lim}}}, \tag{3}$$

$$\sigma_1 + \varepsilon \left(\lambda_1^{\text{pre}} \lambda_2 \frac{\Phi}{H}\right)^2 = \frac{\mu \left(\left(\lambda_1^{\text{pre}}\right)^2 - \left(\lambda_1^{\text{pre}} \lambda_2\right)^{-2}\right)}{1 - \left(\left(\lambda_1^{\text{pre}}\right)^2 + \lambda_2^2 + \left(\lambda_1^{\text{pre}} \lambda_2\right)^{-2} - 3\right)/J_{\text{lim}}}, \tag{4}$$

where Gent material model is used with μ being the shear modulus and J_{lim} being the limiting stretch [25]. Three sets of experiments with different pre-stretches fit remarkably well with the theory, as shown in Fig. 5, where only one set of fitting parameters are used $\mu = 50$ kPa, $J_{\text{lim}} = 190$.

3. Artificial bicipital muscle actuated by FCDEA

3.1. Design of the bioinspired structure

Human elbow is bent by the extension and contraction of the bicipital muscle and tricipital muscle of the arm. This movement is highly efficient and flexible. Hard robots can mimic the similar function by using gear motor, which is, however, of high cost, large noise and less flexibility. In this work, we design a structure to mimic bicipital muscle using the presented FCDEA. As shown in Fig. 6(a), when we bend the elbow, biceps are contracted and triceps are relaxed. Subject to neutral stimuli, biceps extend and triceps contract, as shown in Fig. 6(b). This process can be simulated by applying a voltage to actuate a FCDEA. The design of the structure is as follows.

The entire structure consists of a forearm, an upper arm and an elbow, as shown in the schematics of Fig. 7. The elbow component consists of a pulley, a counterweight, and a forearm connector. The roller bearing is used to connect the forearm and the upper arm. With consideration of the output momentum and maximum rotation angle, we choose POM plastic bearing 609. The counterweight is used to cancel part of the momentum of the forearm to the rotating shaft. The component of upper arm consists of a backbone plate, a FCDEA, an antagonistic spring, a pulley and a shell to package the structure. The backbone plate is a thin rectangular plate with length of 220 mm and width of 50 mm. The FCDEA with original length of 10 mm and width of 10 mm is pre-stretched by $\lambda_1^{\text{pre}} = 5$ and $\lambda_2^{\text{pre}} = 2.9$. The spring with original length of 50 mm is pre-stretched to 100 mm. The pre-stretched FCDEA and the spring are placed on each side of the backbone plate and connected to each other by a rigid wire through the pulley. The volume of the FCDEA is 0.38 cm^3, and the mass is about 5 g. Conducting wires are connected to the FCDEA through the holes in the end cap. A light-weight forearm made of plastic foam is connected to the upper arm through the elbow component.

3.2. Experiment and theoretical modeling

Initially both the FCDEA and the spring are pre-stretched. When actuated by voltage, the FCDEA elongated and correspondingly the spring contracts and the forearm rotates along the shaft at the elbow component, as shown in Fig. 8. Appropriate magnitude of pre-stretch is introduced to maintain the spring in tension during

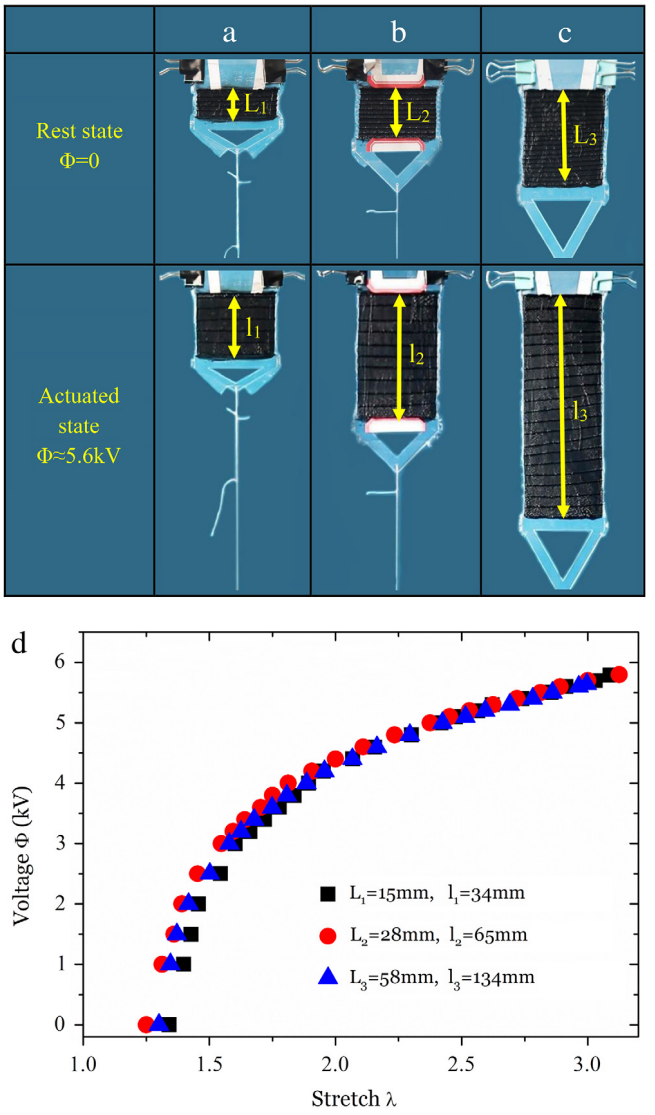


Fig. 4. The presented FCDEA is independent of length ratio. The original length is (a) 15 mm (b) 28 mm (c) 58 mm. (d) The experimental data of voltage versus actuation strain along vertical direction.

Table 1
Comparisons on the unidirectional actuation strain of dielectric elastomer actuators using fiber constraints.

	Dielectric material	Fiber material	Maximum strain (%)
Lu et al. [18]	VHB 4905, 4910	Nylon fiber, carbon fiber	25, 28
Huang et al. [21]	VHB 4905	Nylon fiber	36
Cakmak et al. [22]	VHB 4905	Carbon nanotube	55
Bolzmacher et al. [23]	CF 19-2186	Nylon fiber	28
The present work	VHB 4905	Glass fiber	142

actuation. By adjusting the applied voltage, the position of the rotating forearm can be controlled, neglecting the viscous effect and inertia effect. In the current experiment, a ramp voltage of 30 V/s is applied to the structure until electrical breakdown. Taking the rest state without voltage as reference, up to 70° of rotation has been achieved. The deformation history of voltage-induced rotation is shown in Fig. 8(d), where the symbols represent the experimental data and the solid curve is the theoretical modeling. As shown in the schematics of Fig. 8(c), the geometry relates the change of displacement to the rotation angle and the stretch of the FCDEA.

$$x - x_0 = R\,(\theta - \theta_0) = \left(\lambda_2 - \lambda_2^{\text{pre}}\right) L_2, \tag{5}$$

where $\theta_0 = 28°$, and $R = 23$ mm. The force balance of the spring is simply assumed to be a linear relation $F_s = k\,(x - x_0)$, where k is the stiffness of the spring as a fitting parameter. The force balance of the FCDEA is presented by

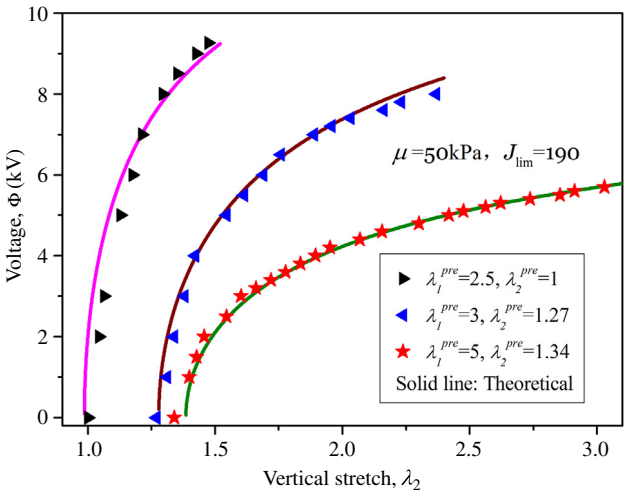


Fig. 5. Comparisons between theory and experiment on the actuation behavior of FCDEA with different pre-stretches.

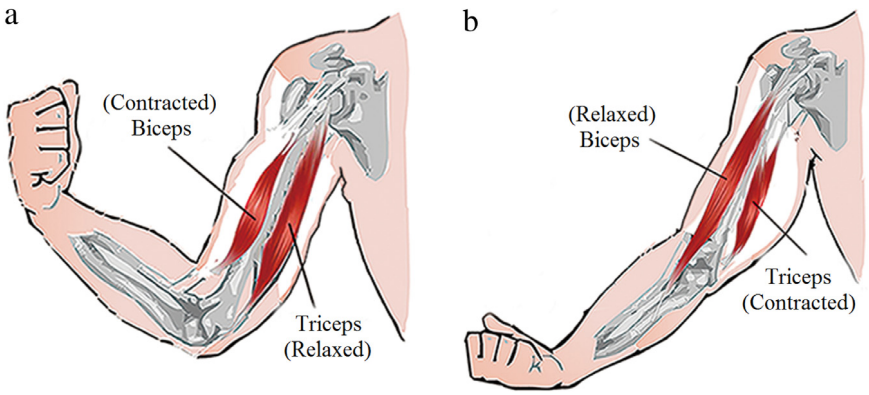


Fig. 6. Schematics of actuation behavior of human muscles. The bicipital muscle is (a) contracted, (b) relaxed.

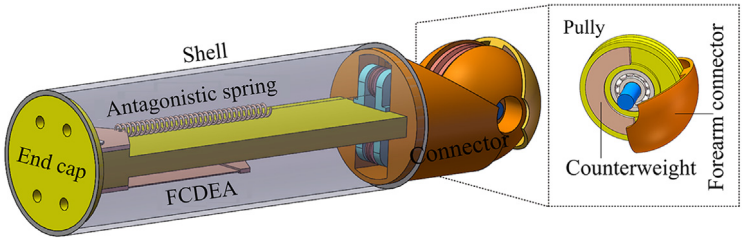


Fig. 7. Components of the designed artificial arm: forearm, upper arm and the elbow.

Eq. (3). The equilibrium of momentum relates the force of the spring F_s and the force of the FCDEA F_{DEA} as

$$(F_s - F_{\mathrm{DEA}})\, R = (m_1 g b_1 - m_2 g b_2)\cos\theta \tag{6}$$

where $m_1 = 4$ g, $m_2 = 17.5$ g, $b_1 = 90$ mm, and $b_2 = 21$ mm. Summing up the equations above and using the same set of material parameters in Fig. 5, we plot the voltage–rotation curve in Fig. 8(d). The agreement with the experimental data is fine except for small deviations under high voltage level. During the process of voltage-induced rotation, the stretch changes from 2.9 to 5 and the rotation angle changes from 28° to 98°.

The presented bioinspired structure without any optimization can output large displacement subject to a voltage. The response time of the actuator subject to a step electrical stimuli is within a second. The loading signal can be programmed according to requirements. Moreover, feedback control can be incorporated to the structure to realize more complex functions. However, the viscous nature of the currently used material VHB 4905 lowers the

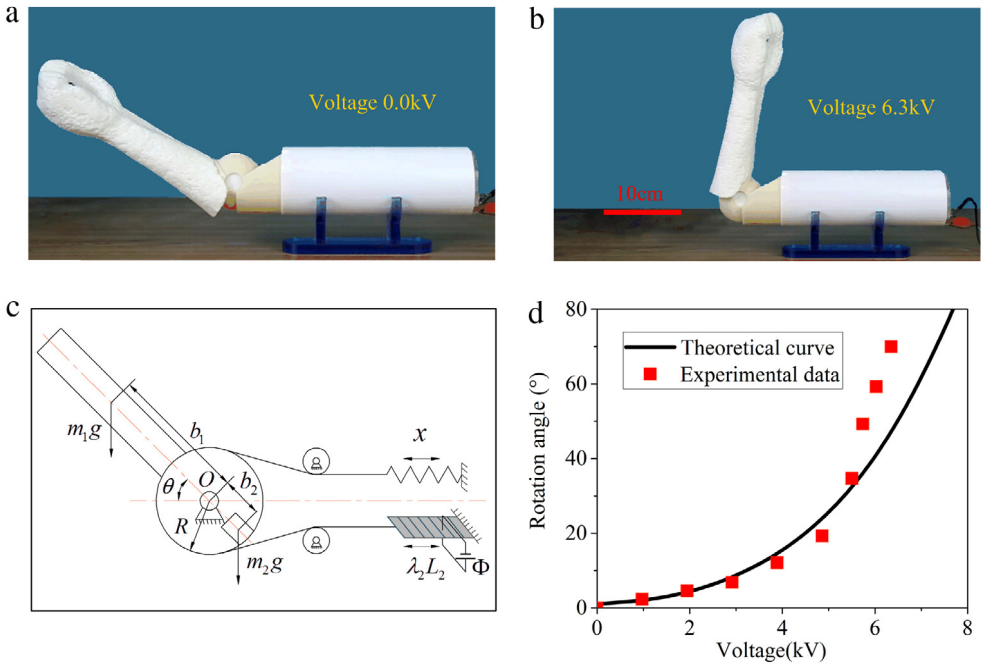


Fig. 8. Under an applied voltage, the forearm rotates 70° relevant to the upper arm. (a) Rest state. (b) Actuated state. (c) The experimental data of voltage–rotation with an exponential fit.

performance on precise control and cyclic loading. The mechanisms applying the high voltage consumes large volume and mass, making the entire system less efficient. A more compact high voltage amplifier and a new kind of dielectric elastomer with higher electrical permittivity and less viscosity will make the presented design more practically useful in the field of artificial muscle.

4. Conclusions

A design of fiber constrained dielectric elastomer actuator is presented. Suitable stiff fibers are selected to withstand large horizontal pre-stretch. Under an applied voltage, up to 142% unidirectional actuation strain is obtained along the vertical direction. The presented design is independent of length ratio, making it promising to mimic the natural muscle of slender shape. A theoretical model to predict the voltage–deformation relation based on ideal dielectric elastomer model agrees with the experimental results very well. A bioinspired artificial bicipital arm consisting of upper arm, elbow and forearm is designed by incorporating the presented FCDEA. A voltage induced rotation of 70° for the forearm relative to the fixed upper arm demonstrates the capability of large actuated deformation. The voltage–rotation curve is also compared to the theoretical modeling. The presented actuator with its unique attribute of large unidirectional deformation would contribute to the development of soft machines made of electroactive polymers.

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